



2014

TRIAL

HIGHER SCHOOL CERTIFICATE

EXAMINATION

GIRRAWEEEN HIGH SCHOOL

MATHEMATICS EXTENSION 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board – approved calculators may be used
- A table of standard integrals is provided
- Show all necessary working in

Questions 11-16

Total marks - 100

Section 1 pages 2-4

10 marks

- Attempt Questions 1-10
- Allow about 20 minutes for this section

Section 2 pages 5 - 13

- Attempt Questions 11 - 16
- Allow about 2 hours 40 minutes for this section

SECTION 1

Multiple Choice (10 marks) Circle your answer on the question paper.

1. For any complex number z , $\arg(z) + \arg(\bar{z})$ equals

(A) 0 (B) $n\pi$ (C) $-n\pi$ (D) $\frac{\pi}{4}$

2. The conjugate of $\frac{1+2i}{3-i}$ is

(A) $1+7i$ (B) $1-7i$ (C) $\frac{1+7i}{10}$ (D) $\frac{1-7i}{10}$

3. The roots of $z^3 - 1 = 0$ in modulus argument form are

(A) $\text{cis}0, \text{cis}\frac{2\pi}{3}$ (C) $\text{cis}0, \text{cis}\frac{\pi}{3}$
 (B) $\text{cis}0, \text{cis}\frac{2\pi}{3}, \text{cis}\left(-\frac{2\pi}{3}\right)$ (D) $\text{cis}0, \text{cis}\frac{\pi}{3}, \text{cis}\left(-\frac{\pi}{3}\right)$

4. For the ellipse with equation $\frac{x^2}{16} + \frac{y^2}{36} = 1$, what is the eccentricity?

(A) $\frac{2\sqrt{7}}{6}$ (B) $\frac{2\sqrt{11}}{6}$ (C) $\frac{2\sqrt{5}}{6}$ (D) $\frac{2\sqrt{3}}{6}$

5. For the hyperbola $\frac{(x+2)^2}{9} - \frac{(y-1)^2}{16} = 1$, the coordinates of one of the focus is

(A) (7,1) (B) (-7, -1) (C) (3,1) (D) (-3, 1)

6. The value of $\int_0^1 x(1-x)^{99} dx$ is

(A) $\frac{1}{10010}$ (B) $\frac{1}{10100}$ (C) $\frac{11}{10100}$ (D) $\frac{11}{10010}$

7.

Reduce into partial fractions: $\frac{3x+1}{(x-2)^2(x+2)}$

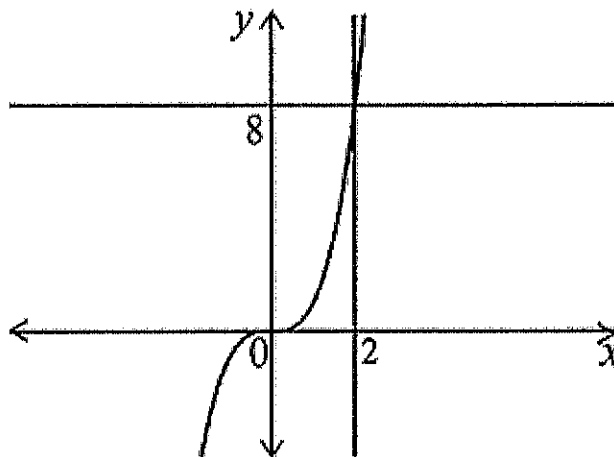
(A) $\frac{5}{16(x-2)} + \frac{7}{4(x-2)^2} - \frac{5}{16(x+2)}$

(B) $\frac{5}{16(x-2)} - \frac{7}{4(x-2)^2} + \frac{5}{16(x+2)}$

(C) $\frac{5}{16(x-2)} - \frac{7}{4(x-2)^2} - \frac{5}{16(x+2)}$

(D) $\frac{5}{16(x-2)} + \frac{7}{4(x-2)^2} + \frac{5}{16(x+2)}$

8. The volume of the solid generated when the region bounded by $y = x^3$, $x = 0$, $y = 8$ is revolved about the line $x = 2$.



(A) $\frac{963\pi}{5}$

(B) $\frac{144\pi}{5}$

(C) $\frac{153\pi}{5}$

(D) $\frac{320\pi}{5}$

9. What is the angle at which a road must be banked so that a car may round a curve with a radius of 200 metres at 100 km/h without sliding. Assume that the road is smooth.

- (A) 24.49° (B) 23.49° (C) 22.49° (D) 21.49°

10. The polynomial $P(x) = x^4 + ax^2 + bx + 28$ has a double root at $x = 2$. What are the values of a and b ?

- (A) $a = -11$ and $b = -12$ (C) $a = -11$ and $b = 12$
(B) $a = -5$ and $b = -12$ (D) $a = -5$ and $b = 12$

END OF SECTION 1

Question 11 (15 marks)**Marks**

(a) $\int_0^{\ln 2} \frac{e^x dx}{1+e^x}$ 2

(b) $\int \frac{dx}{3x^2 + 6x + 10}$ 2

(c) $\int_0^{\frac{\pi}{3}} \frac{dx}{5 - 4 \cos x}$ 3

(d) (i) If $I_n = \int_{-1}^0 x^n \sqrt{1+x} dx$ for $n = 0, 1, 2, \dots$ show that $I_n = -\frac{2n}{2n+3} I_{n-1}$

for $n = 1, 2, 3, \dots$ 3

(ii) Hence evaluate $\int_{-1}^0 x^2 \sqrt{1+x} dx$ 2

(e) Use the result $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ 3

Examination continues on next page

Question 12 (15 marks)

(a) If $z = \sqrt{3} + i$ and $w = 1 - i$

(i) Write $\frac{z}{w}$ in the form $a + ib$ where a and b are real numbers. 1

(ii) write $\frac{z}{w}$ in mod - arg form. 2

(iii) Hence find the exact values of $\sin \frac{5\pi}{12}$ and $\cos \frac{5\pi}{12}$ 2

(b) Sketch the following on separate Argand diagrams.

(i) $\arg(z - (-1 + 2i)) = -\frac{2\pi}{3}$ 2

(ii) $2 < |z| \leq 4$ and $-\frac{\pi}{4} < \arg(z) \leq \frac{\pi}{2}$ 2

(c) If n is a positive integer, prove that $(\sqrt{3} + i)^n + (\sqrt{3} - i)^n = 2^{n+1} \cos \frac{n\pi}{6}$ 2

(d) z satisfies $|z - 2i| = 1$, and the point P represents z on an Argand diagram.

(i) Sketch the locus of P as z varies. 1

(ii) Find the maximum and minimum values of $\arg z$, where $-\pi < \arg z \leq \pi$. 2

(iii) Find the value of z when $\arg z$ takes the maximum value, and express in modulus-argument form. 1

Examination continues on next page

Question 13 (15 marks)

- (a) If $2 - 3i$ is a zero of the polynomial $z^3 + pz + q$ where p and q are real, find the values of p and q . 2
- (b) If α, β and γ are the roots of the equation $x^3 + 6x + 1 = 0$, find the polynomial equation whose roots are $\alpha\beta, \alpha\gamma$ and $\beta\gamma$. 3
- (c) Given that the quartic polynomial $P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$ has a zero of multiplicity 3, factorise $P(x)$ completely and find all its zeros. 3
- (d) (i) Solve $z^8 + 1 = 0$. Leave answers in mod- arg form. 2
- (ii) Factorise $z^8 + 1$ into real quadratic factors. 2
- (iii) Show that $\cos 4\theta = 8 \left(\cos^2 \theta - \cos^2 \frac{\pi}{8} \right) \left(\cos^2 \theta - \cos^2 \frac{3\pi}{8} \right)$ 3

Examination continues on next page

Question 14 (15 marks)

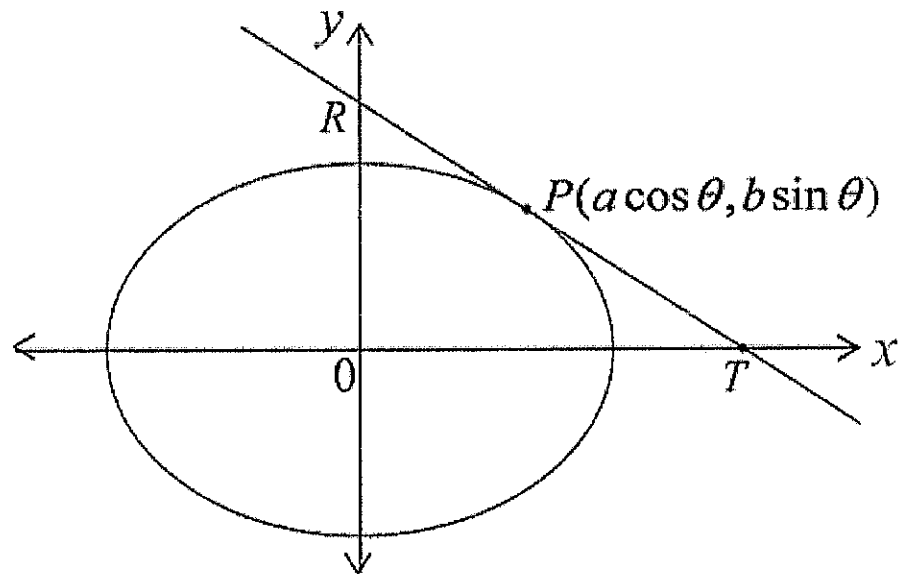
- (a) The ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has a tangent at the point $P(a \cos \theta, b \sin \theta)$.

The tangent cuts the x - axis at T and the y - axis at R .

- (i) Show that the equation of the tangent at the point P is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

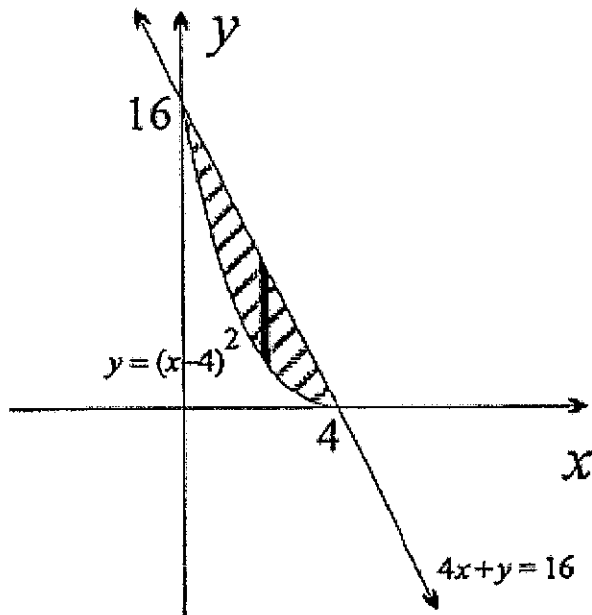
3



- (ii) If T is the point of intersection between the tangent at point P and one of the directrices of the ellipse, show that $\cos \theta = e$ 3
- (iii) Hence find the angle that the focal chord through P makes with the x - axis. 1
- (iv) Using similar triangles or otherwise show that $RP = e^2 RT$. 3

Examination continues on next page

(b)



The region enclosed by the curve $y = (x-4)^2$ and the line $4x + y = 16$ is shaded in the diagram above. A solid is formed with this region as its base. When the solid is sliced perpendicular to the x -axis, each cross-section is an equilateral triangle with its base in the xy -plane.

- (i) Show that the area of the cross-section x units to the right of the y -axis is

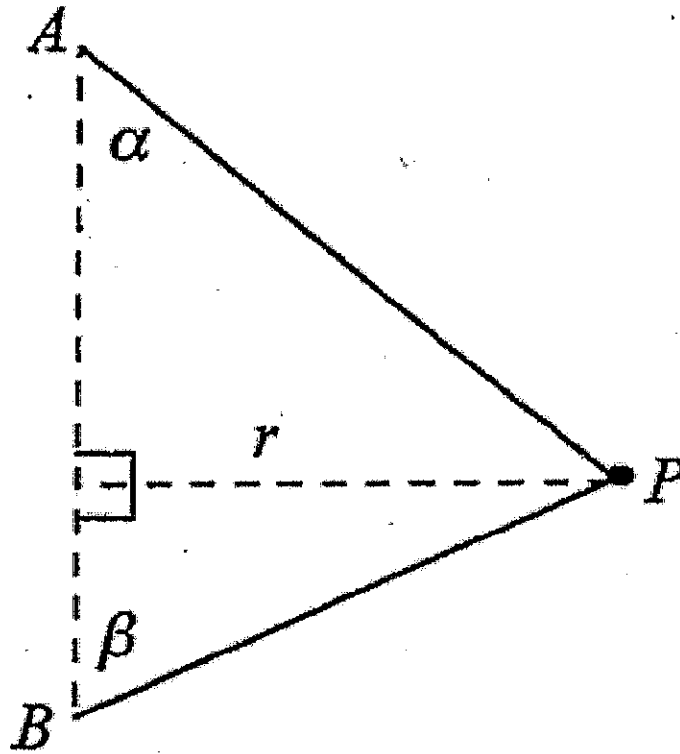
$$\frac{\sqrt{3}x^2(4-x)^2}{4}, \text{ where } 0 \leq x \leq 4. \quad 2$$

- (ii) Hence find the volume of the solid. 3

Examination continues on next page

Question 15 (15 marks)

- (a) A and B are two fixed points with B vertically below A . P is a particle with mass m kg. Two strings with ends fixed at A and B are fastened to P . Particle P moves in a horizontal circle of radius r metres with a constant angular velocity ω radians per second so that both strings remain taut, making angles α, β respectively with vertical. The tension in the string AP and BP are T_1 Newtons and T_2 Newtons respectively. The acceleration due to gravity is $g \text{ m/s}^2$.



- (i) Draw a diagram showing the forces acting on the particle P . 1
- (ii) By resolving the forces, find an expression for the tension in each part of the string in terms of m, r, ω^2 and $\sin(\alpha + \beta)$. 5
- (iii) Show that $\omega > \sqrt{\frac{g \tan \alpha}{r}}$ for both strings to be taut. 1

Examination continues on next page

(b) A body of mass m in falling from rest, experiences air resistance of magnitude kv^2 per unit mass, where k is a positive constant.

- (i) Write the equation of motion of the body and find the value of the terminal velocity V of the body in terms of k and g (acceleration due to gravity)

[Take $g = 9.8m/s^2$] 2

- (ii) If w is the velocity of the body when it reaches the ground, show that the

distance S fallen is given by $S = -\frac{1}{2k} \ln \left(1 - \frac{w^2}{V^2} \right)$. 3

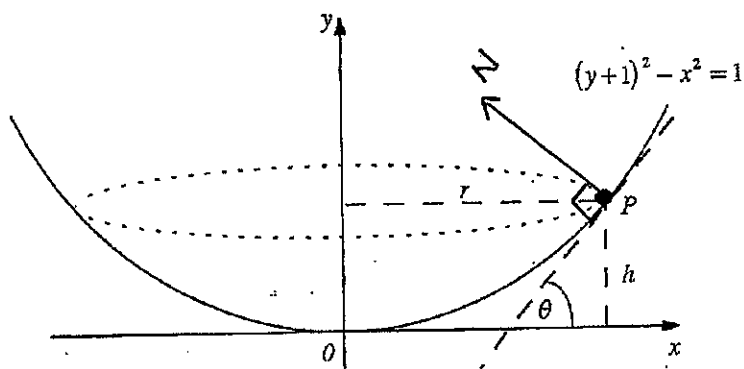
- (iii) With air resistance remaining the same, prove that if the body is projected vertically upwards from the ground with velocity U , then it will attain its

greatest height H where $H = \frac{1}{2k} \ln \left(1 + \frac{U^2}{V^2} \right)$, and return to the ground with

velocity w given by $w^2 = U^2 + V^2$. 3

Examination continues on next page

Question 16 (15 marks)



- (a) A smooth bowl is formed by rotating the hyperbola $(y+1)^2 - x^2 = 1$ around the y -axis. A particle P of mass m kg travels around the inside of the bowl with constant angular velocity ω radians per second in a horizontal circle of radius r metres at a height h metres above the bottom of the bowl.

- (i) Show that if the tangent to the hyperbola $(y+1)^2 - x^2 = 1$ at the point

$$(x_1, y_1) \text{ makes an angle } \theta \text{ with the positive } x\text{-axis, then } \tan \theta = \frac{x_1}{1+y_1} \cdot 2$$

- (ii) Copy the diagram and write the equations of motion by resolving forces

on P . 2

- (iii) Show that $\omega^2 = \frac{g}{1+h}$. 2

- (iv) Show that the force N Newtons exerted by the particle P on the bowl is given

$$\text{by } N = mg \sqrt{2 - \frac{1}{(1+h)^2}}. \quad 2$$

- (v) If the linear speed of the particle is $\sqrt{\frac{3g}{2}}$ m/s, find h and the force exerted

by the particle on the bowl. 2

Examination continues on next page

- (b) (i) A truck of mass 5 tonne is travelling around a circular section of the highway which is banked at an angle $\theta = \tan^{-1}\left(\frac{1}{5}\right)$. The radius of the curve is 450 m. Find the speed of the truck, so that there is no lateral friction on the tyres. (Take $g = 10 \text{ m/s}^2$) 2
- (ii) Find the sideways frictional force between the tyres and the road for the truck of mass 5 tonne, if the speed of the truck is 126 km/h. 3

END OF EXAMINATION

Extension 2 Total HSC 2014 - solutions

Section 1

1. (A)

$$2. \frac{1+2i}{3-i} \times \frac{3+i}{3+i}$$

$$= \frac{3+i+6i+2i^2}{3^2-i^2}$$

$$= \frac{3+7i-2}{9+1}$$

$$= \frac{1+7i}{10} \quad \text{(D)}$$

3. (B)

$$4. \frac{x^2}{16} + \frac{y^2}{36} = 1$$

$$c = \sqrt{1 - \frac{a^2}{b^2}}$$

$$= \sqrt{1 - \frac{16}{36}}$$

$$= \sqrt{\frac{20}{36}}$$

$$= \frac{2\sqrt{5}}{6} \quad \text{(C)}$$

5. Four are

$$(h \pm ae, k)$$

$$= (-2 \pm 3 \times \frac{5}{3}, 1) \quad \text{(C)}$$

$$= (3, 1) \text{ and } (-7, 1)$$

$$6. \int_0^1 x(1-x)^{99} dx$$

$$= x \times \frac{(1-x)^{100}}{-100} \Big|_0^1 - \int_0^1 1 \times \frac{(1-x)^{100}}{-100} dx$$

$$= 0 + \frac{1}{100} \int_0^1 (1-x)^{100} dx$$

$$= \frac{1}{100} \left[\frac{(1-x)^{101}}{-101} \right]_0^1$$

$$= \frac{-1}{100} \times \frac{1}{101} \left[(1-x)^{101} \right]_0^1$$

$$= \frac{-1}{100 \times 101} (0 - 1) = \frac{1}{10100} \quad \text{(B)}$$

7. (A)

$$8. R = 2 \quad r = 2 - x$$

$$R + r = 4 - x \quad ; \quad R - r = x$$

$$\text{Area} = \pi(R+r)(R-r)$$

$$= \pi(4-x)x$$

$$= \pi(4x - x^2)$$

$$\Delta V = \pi(4x - x^2) \Delta y$$

$$V = \pi \int_0^8 (4y^{\frac{1}{3}} - y^{\frac{2}{3}}) dy$$

$$= \pi \left[4 \frac{y^{\frac{4}{3}}}{\frac{4}{3}} - \frac{y^{\frac{5}{3}}}{\frac{5}{3}} \right]_0^8$$

$$y = x^3$$

$$x = y^{\frac{1}{3}}$$

$$x^2 = y^{\frac{2}{3}}$$

$$= \pi \left[4y^{\frac{4}{3}} \times \frac{3}{4} - y^{\frac{5}{3}} \times \frac{3}{5} \right]_0^8$$

$$= \pi \left[3y^{\frac{4}{3}} - \frac{3}{5} y^{\frac{5}{3}} \right]_0^8$$

$$= \pi \left(3 \times 8^{\frac{4}{3}} - \frac{3}{5} \times 8^{\frac{5}{3}} \right) - 0$$

$$= \pi \left(3 \times 16 - \frac{3}{5} \times 32 \right)$$

$$= \frac{144\pi}{5} \quad (B)$$

$$9. \text{ time} = \frac{V^2}{rg}$$

$$V = 100 \text{ km/h}$$

$$= \frac{1000}{36} \text{ m/s}$$

$$\text{time} = \frac{1000 \times 1000}{36 \times 36 \times 200 \times 9.8}$$

$$\theta = 21.49^\circ \quad (D)$$

$$10. p(2) = 0 \text{ and } p'(2) = 0$$

$$p(2) = 16 + 4a + 2b + 28$$

$$4a + 2b = -44$$

$$2a + b = -22$$

$$p'(2) = 4x^3 + 2ax + b$$

$$p'(2) = 4 \times 8 + 4a + b$$

$$= 32 + 4a + b$$

$$2a + b = -22$$

$$4a + b = -32$$

$$2a = -10 \quad a = -5$$

$$b = -22 - 2a$$

$$= -22 + 10 = -12$$

(B)

1A

6B

2D

7A

3B

8B

4C

9D

5C

10B

Question 11

$$(a) \int_0^{\ln 2} \frac{e^x dx}{1+e^x}$$

$$\text{Let } u = 1 + e^x$$

$$\frac{du}{dx} = e^x$$

$$e^x dx = du$$

$$\text{When } x=0, u = 1 + e^0 = 2$$

$$\text{When } x = \ln 2, u = 1 + e^{\ln 2} = 3$$

$$I = \int_2^3 \frac{du}{u} = [\log u]_2^3$$

$$= \log 3 - \log 2$$

$$= \log \frac{3}{2}$$

$$(b) \int \frac{dx}{3x^2 + 6x + 10}$$

$$3x^2 + 6x + 10$$

$$= 3(x^2 + 2x + \frac{10}{3})$$

$$= 3(x^2 + 2x + 1 - 1 + \frac{10}{3})$$

$$= 3((x+1)^2 + \frac{10-3}{3})$$

$$= 3((x+1)^2 + \frac{7}{3})$$

$$\int \frac{dx}{3x^2 + 6x + 10} = \int \frac{dx}{3((x+1)^2 + \frac{7}{3})}$$

$$= \frac{1}{3} \int \frac{dx}{(x+1)^2 + (\frac{\sqrt{7}}{\sqrt{3}})^2}$$

$$= \frac{1}{3} \times \frac{1}{\frac{\sqrt{7}}{\sqrt{3}}} \tan^{-1} \left(\frac{x+1}{\frac{\sqrt{7}}{\sqrt{3}}} \right) + C$$

$$= \frac{1}{3} \times \frac{\sqrt{3}}{\sqrt{7}} \tan^{-1} \left(\frac{\sqrt{3}(x+1)}{\sqrt{7}} \right)$$

$$= \frac{1}{\sqrt{21}} \tan^{-1} \left(\frac{\sqrt{3}(x+1)}{\sqrt{7}} \right)$$

$$(c) \int_0^{\frac{\pi}{3}} \frac{dx}{5 - 4 \cos x}$$

$$t = \tan \frac{x}{2} ; \frac{dt}{dx} = \left(\sec^2 \frac{x}{2} \right) \frac{1}{2}$$

$$= \frac{1}{2} \sec^2 \frac{x}{2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$dt = \frac{1}{2} (1 + \tan^2 \frac{x}{2}) dx$$

$$dt = \frac{1}{2} (1 + t^2) dx$$

$$dx = \frac{2 dt}{1 + t^2}$$

when $x=0$, $t = \tan \frac{0}{2} = 0$

when $x = \frac{\pi}{3}$, $t = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$

$$5 - 4 \cos x = 5 - 4 \left(\frac{1-t^2}{1+t^2} \right)$$

$$= \frac{5(1+t^2) - 4(1-t^2)}{1+t^2}$$

$$= \frac{5+5t^2-4+4t^2}{1+t^2}$$

$$= \frac{1+9t^2}{1+t^2}$$

$$I = \int_0^{\frac{1}{\sqrt{3}}} \frac{1+t^2}{1+9t^2} \times \frac{2dt}{1+t^2}$$

$$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{9(t^2 + \frac{1}{9})}$$

$$= \frac{2}{9} \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{t^2 + (\frac{1}{3})^2}$$

$$= \left[\frac{2}{9} \times \frac{1}{\frac{1}{3}} \tan^{-1} \left(\frac{t}{\frac{1}{3}} \right) \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= \frac{2}{9} \times 3 \left[\tan^{-1} 3t \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= \frac{2}{3} \left(\tan^{-1} 3 \times \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right)$$

page 4

$$= \frac{2}{3} \left(\tan^{-1} \sqrt{3} - \tan^{-1}(0) \right)$$

$$= \frac{2}{3} \left(\frac{\pi}{3} - 0 \right) = \underline{\underline{\frac{2\pi}{9}}}$$

$$(d) I_n = \int_{-1}^0 x^n \sqrt{1+x} dx$$

$$= \frac{2}{3} \left[x^n (1+x)^{\frac{3}{2}} \right]_{-1}^0$$

$$- \frac{2}{3} \int_{-1}^0 n x^{n-1} (1+x) \sqrt{1+x} dx$$

$$= 0 - \frac{2n}{3} \left\{ \int_{-1}^0 x^{n-1} \sqrt{1+x} dx + \int_{-1}^0 x^n \sqrt{1+x} dx \right\}$$

$$I_n = -\frac{2n}{3} I_{n-1} - \frac{2n}{3} I_n$$

$$\left(1 + \frac{2n}{3} \right) I_n = -\frac{2n}{3} I_{n-1}$$

$$(3+2n) I_n = -2n I_{n-1}$$

$$\underline{\underline{I_n = \frac{-2n}{2n+3} I_{n-1}}}$$

$$(ii) \int_{-1}^0 \sqrt{1+x} \, dx$$

$$= \frac{2}{3} \left[(1+x)^{\frac{3}{2}} \right]_{-1}^0$$

$$= \frac{2}{3} (1-0) = \frac{2}{3}$$

$$I_2 = -\frac{4}{7} I_1$$

$$= \left(-\frac{4}{7}\right) \left(-\frac{2}{5}\right) I_0$$

$$= -\frac{4}{7} \times -\frac{2}{5} \times \frac{2}{3}$$

$$= \frac{16}{105}$$

$$(e) \text{ Let } I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \, dx \quad \text{--- (1)}$$

$$\text{Also } I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin(\frac{\pi}{2}-x)}}{\sqrt{\sin(\frac{\pi}{2}-x)} + \sqrt{\cos(\frac{\pi}{2}-x)}} \, dx \quad \left(\because \int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx \right)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} \, dx \quad \text{--- (2)} \quad \left(\because \begin{aligned} \sin(90^\circ - \alpha) &= \cos \alpha \\ \cos(90^\circ - \alpha) &= \sin \alpha \end{aligned} \right)$$

① + ② gives

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} \, dx = \int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2}$$

Question 12

(a) (i) $z = \sqrt{3} + i, w = 1 - i$

$$\frac{z}{w} = \frac{\sqrt{3} + i}{1 - i} \times \frac{1 + i}{1 + i}$$

$$= \frac{(\sqrt{3} + i)(1 + i)}{(1 - i)(1 + i)}$$

$$= \frac{\sqrt{3} + \sqrt{3}i + i + i^2}{1 - i^2}$$

$$= \frac{\sqrt{3} - 1 + i(\sqrt{3} + 1)}{2}$$

$$= \frac{\sqrt{3} - 1}{2} + i \frac{(\sqrt{3} + 1)}{2}$$

(ii) $z = \sqrt{3} + i$

$$|z| = \sqrt{3 + 1} = 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}} \quad \alpha = \frac{\pi}{6}$$

$$\alpha = \frac{\pi}{6}$$

$$z = 2 \operatorname{cis} \frac{\pi}{6}$$

$w = 1 - i$

$$|w| = \sqrt{1 + 1} = \sqrt{2}$$

$$\tan \alpha = 1 \quad \alpha = \frac{\pi}{4}$$

$$\alpha = -\frac{\pi}{4}$$

$$w = \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4}\right)$$

page 6

$$\frac{z}{w} = \frac{2 \operatorname{cis} \frac{\pi}{6}}{\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4}\right)}$$

$$= \frac{2}{\sqrt{2}} \operatorname{cis} \left(\frac{\pi}{6} + \frac{\pi}{4}\right)$$

$$= \sqrt{2} \operatorname{cis} \frac{5\pi}{12}$$

(ii) $\sqrt{2} \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right)$

$$= \frac{\sqrt{3} - 1}{2} + i \frac{(\sqrt{3} + 1)}{2}$$

Equating real and imaginary parts we get-

$$\cos \frac{5\pi}{12} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

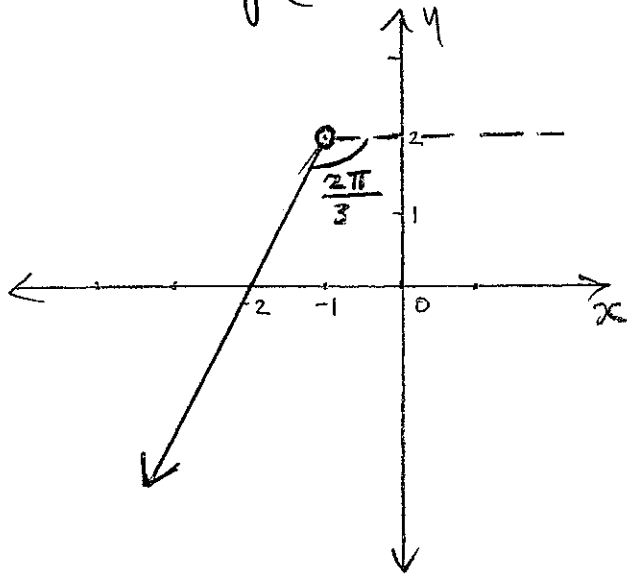
$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} - \sqrt{2}}{4}$$

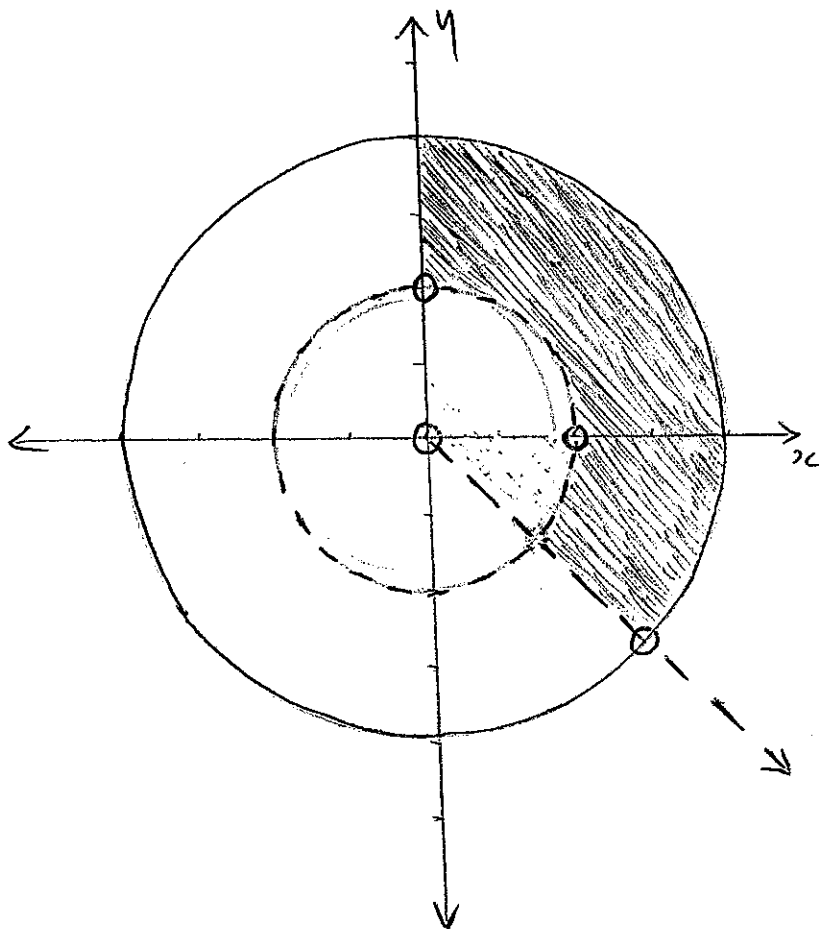
$$\sin \frac{5\pi}{12} = \frac{\sqrt{3} + 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

(b) (i) $\arg(z - (-1+2i)) = -\frac{2\pi}{3}$



(ii) $2 < |z| \leq 4$ and $-\frac{\pi}{4} < \arg z \leq \frac{\pi}{2}$



(c) $\sqrt{3}+i = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$

$(\sqrt{3}+i)^n = 2^n \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)^n$

$= 2^n \left(\cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6} \right)$ by De Moivre's theorem

Page 17

$\sqrt{3}-i = 2 \left(\cos \frac{-\pi}{6} + i \sin \frac{-\pi}{6} \right)$

$= 2 \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$

$(\because \sin(-\theta) = -\sin \theta)$
 $\cos(-\theta) = \cos \theta$

$(\sqrt{3}-i)^n = 2^n \left(\cos \frac{n\pi}{6} - i \sin \frac{n\pi}{6} \right)$

(by De Moivre's theorem)

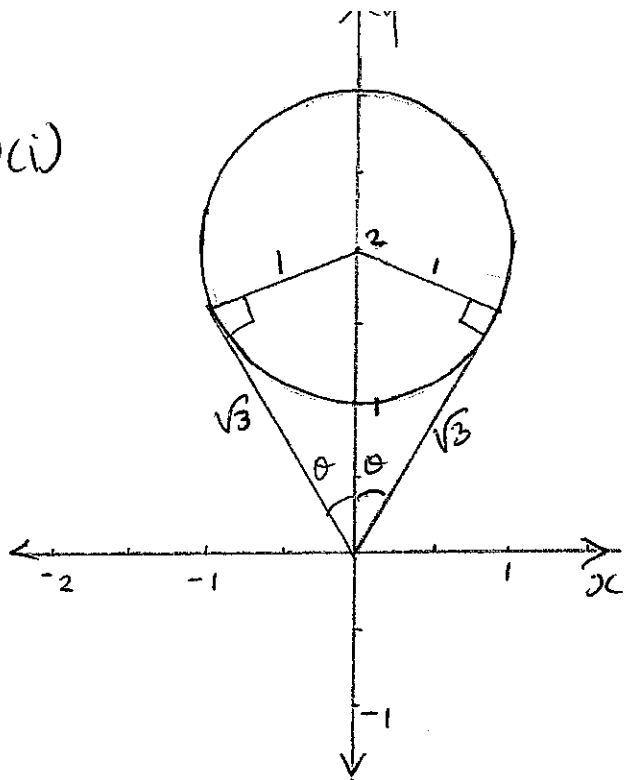
$(\sqrt{3}+i)^n + (\sqrt{3}-i)^n$

$= 2^n \left(\cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6} + \cos \frac{n\pi}{6} - i \sin \frac{n\pi}{6} \right)$

$= 2^n \times 2 \cos \frac{n\pi}{6}$

$= \underline{\underline{2^{n+1} \cos \frac{n\pi}{6}}}$

(d)(i)



$$(ii) \tan \theta = \frac{1}{\sqrt{3}} \quad \theta = \frac{\pi}{6}$$

$$\text{minimum } \arg(z) = \frac{\pi}{2} - \frac{\pi}{6}$$

$$= \frac{\pi}{3}$$

$$\text{Maximum } \arg(z) = \frac{\pi}{2} + \frac{\pi}{6}$$

$$= \frac{2\pi}{3}$$

$$(iii) z = \sqrt{3} \operatorname{cis} \frac{2\pi}{3}$$

Question 13

$$(a) z^3 + pz + q = 0$$

$$(2-3i)^3 + (2-3i)p + q = 0$$

$$(2-3i)^3 = (2-3i)^2(2-3i)$$

$$= (-5-12i)(2-3i)$$

$$= -10 + 15i - 24i + 36i^2$$

$$= -10 - 9i - 36$$

$$= -46 - 9i$$

page 8

$$-46 - 9i + (2-3i)p + q = 0$$

$$-46 - 9i + 2p - i3p + q = 0$$

$$-46 + 2p + q - 9i - 3pi = 0$$

Equating real and imaginary parts

$$-46 + 2p + q = 0$$

$$2p + q = 46 \quad \text{--- (1)}$$

$$-9 - 3p = 0 \quad \text{--- (2)}$$

$$\text{From (2)} \quad 3p = -9$$

$$p = -3$$

Substitute (2) in (1)

$$q = 46 - 2p$$

$$= 46 - 2(-3)$$

$$= 52$$

$$\underline{\underline{p = -3, q = 52}}$$

$$(b) x^3 + 6x + 1 = 0$$

$$\alpha \beta r = -1$$

$$\alpha \beta = \frac{-1}{r}$$

$$\beta r = \frac{-1}{\alpha}$$

$$\alpha r = \frac{-1}{\beta}$$

$$\text{let } y = \frac{-1}{x} \Rightarrow x = \frac{-1}{y}$$

Required polynomial is

$$\left(\frac{-1}{y}\right)^3 + 6\left(\frac{-1}{y}\right) + 1 = 0$$

$$-1 - 6y^2 + y^3 = 0$$

$$y^3 - 6y^2 - 1 = 0$$

$$\underline{\underline{x^3 - 6x^2 - 1 = 0}}$$

$$(c) P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$$

$$P'(x) = 4x^3 - 15x^2 + 18x + 81$$

$$P''(x) = 12x^2 - 30x - 18$$

$$P'''(x) = 0 \Rightarrow 2x^2 - 5x - 3 = 0$$

$$x = 3, \frac{-1}{2}$$

$$P(3) = 3^4 - 5 \times 3^3 - 9 \times 3^2 + 81 \times 3 - 108 = 0$$

$\therefore 3$ is a root of multiplicity 3.

Let β be the fourth root

$$27\beta = -108$$

$$\beta = \frac{-108}{27} = -4$$

$$\underline{\underline{P(x) = (x-3)^3(x+4)}}$$

$$(d) (i) z^8 = 1 = \text{cis } \pi$$

$$= \text{cis}(\pi + 2k\pi) \quad k \in \mathbb{Z}$$

$$z = [\text{cis}(2k+1)\pi]^{\frac{1}{8}}$$

$$= \text{cis} \frac{(2k+1)\pi}{8}, \quad k = 0, 1, 2, 3, \dots, 7$$

$$k=0, z_1 = \text{cis} \frac{\pi}{8}$$

$$k=1, z_2 = \text{cis} \frac{3\pi}{8}$$

$$k=2, z_3 = \text{cis} \frac{5\pi}{8}$$

$$k=3, z_4 = \text{cis} \frac{7\pi}{8}$$

$$k=4, z_5 = \text{cis} \frac{9\pi}{8} = \text{cis} \frac{-7\pi}{8}$$

$$k=5, z_6 = \text{cis} \frac{11\pi}{8} = \text{cis} \frac{-5\pi}{8}$$

$$k=6, z_7 = \text{cis} \frac{13\pi}{8}$$

$$= \text{cis} \frac{-3\pi}{8}$$

$$k=7, z_8 = \text{cis} \frac{15\pi}{8} = \text{cis} \frac{-\pi}{8}$$

$$z^8 + 1 = \left(z - \operatorname{cis} \frac{\pi}{8}\right) \left(z - \operatorname{cis} \frac{-\pi}{8}\right) \left(z - \operatorname{cis} \frac{3\pi}{8}\right) \left(z - \operatorname{cis} \frac{-3\pi}{8}\right)$$

$$\left(z - \operatorname{cis} \frac{5\pi}{8}\right) \left(z - \operatorname{cis} \frac{-5\pi}{8}\right) \left(z - \operatorname{cis} \frac{7\pi}{8}\right) \left(z - \operatorname{cis} \frac{-7\pi}{8}\right)$$

$$\left(z - \operatorname{cis} \frac{\pi}{8}\right) \left(z - \operatorname{cis} \frac{-\pi}{8}\right) = \left(z - \cos \frac{\pi}{8} - i \sin \frac{\pi}{8}\right) \left(z - \cos \frac{\pi}{8} + i \sin \frac{\pi}{8}\right)$$

$$= \left(z - \cos \frac{\pi}{8}\right)^2 - \left(i \sin \frac{\pi}{8}\right)^2$$

$$= z^2 - 2 \cos \frac{\pi}{8} z + \cos^2 \frac{\pi}{8} + \sin^2 \frac{\pi}{8}$$

$$= z^2 - 2 \cos \frac{\pi}{8} z + 1$$

Similarly $\left(z - \operatorname{cis} \frac{3\pi}{8}\right) \left(z - \operatorname{cis} \frac{-3\pi}{8}\right) = z^2 - 2 \cos \frac{3\pi}{8} z + 1$

$$\left(z - \operatorname{cis} \frac{5\pi}{8}\right) \left(z - \operatorname{cis} \frac{-5\pi}{8}\right) = z^2 - 2 \cos \frac{5\pi}{8} z + 1$$

$$\left(z - \operatorname{cis} \frac{7\pi}{8}\right) \left(z - \operatorname{cis} \frac{-7\pi}{8}\right) = z^2 - 2 \cos \frac{7\pi}{8} z + 1$$

$$z^8 + 1 = \left(z^2 - 2 \cos \frac{\pi}{8} z + 1\right) \left(z^2 - 2 \cos \frac{3\pi}{8} z + 1\right) \left(z^2 - 2 \cos \frac{5\pi}{8} z + 1\right) \left(z^2 - 2 \cos \frac{7\pi}{8} z + 1\right)$$

$\div$ by z^4

$$z^4 + \frac{1}{z^4} = \left(z - 2 \cos \frac{\pi}{8} + \frac{1}{z}\right) \left(z - 2 \cos \frac{3\pi}{8} + \frac{1}{z}\right) \left(z - 2 \cos \frac{5\pi}{8} + \frac{1}{z}\right) \left(z - 2 \cos \frac{7\pi}{8} + \frac{1}{z}\right)$$

$$2 \cos 4\theta = \left(2 \cos \theta - 2 \cos \frac{\pi}{8}\right) \left(2 \cos \theta - 2 \cos \frac{3\pi}{8}\right) \left(2 \cos \theta - 2 \cos \frac{5\pi}{8}\right) \left(2 \cos \theta - 2 \cos \frac{7\pi}{8}\right)$$

page 11

$$2 \cos 4\theta = 2^4 \left(\cos \theta - \cos \frac{\pi}{8} \right) \left(\cos \theta - \cos \frac{3\pi}{8} \right) \left(\cos \theta - \cos \frac{5\pi}{8} \right) \left(\cos \theta - \cos \frac{7\pi}{8} \right)$$

divide by 2

$$\cos 4\theta = 8 \left(\cos \theta - \cos \frac{\pi}{8} \right) \left(\cos \theta - \cos \frac{3\pi}{8} \right) \left(\cos \theta - \cos \frac{5\pi}{8} \right) \left(\cos \theta - \cos \frac{7\pi}{8} \right)$$

$$\cos \frac{5\pi}{8} = \cos \left(\pi - \frac{3\pi}{8} \right) = -\cos \frac{3\pi}{8}$$

$$\cos \frac{7\pi}{8} = \cos \left(\pi - \frac{\pi}{8} \right) = -\cos \frac{\pi}{8}$$

$$\therefore \cos 4\theta = 8 \left(\cos \theta - \cos \frac{\pi}{8} \right) \left(\cos \theta - \cos \frac{3\pi}{8} \right) \left(\cos \theta + \cos \frac{3\pi}{8} \right) \left(\cos \theta + \cos \frac{\pi}{8} \right)$$

$$= 8 \left(\cos^2 \theta - \cos^2 \frac{\pi}{8} \right) \left(\cos^2 \theta - \cos^2 \frac{3\pi}{8} \right)$$

Question 14

$$\text{Q (i)} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\frac{dy}{dx} \text{ at } P = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

Equation of tangent

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a \sin \theta y - ab \sin^2 \theta = -b \cos \theta x + ab \cos^2 \theta$$

$$b \cos \theta x + a \sin \theta y = ab(\sin^2 \theta + \cos^2 \theta)$$

$\div$ by ab

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

(ii) At T, $y=0$ and T is a point on the tangent page 12

Substitute $y=0$ in $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

$$\frac{x \cos \theta}{a} = 1 \quad \therefore x = \frac{a}{\cos \theta}$$

$$T\left(\frac{a}{\cos \theta}, 0\right)$$

Equation of director is $x = \frac{a}{e}$

$$\frac{a}{\cos \theta} = \frac{a}{e} \quad \therefore \underline{\underline{\cos \theta = e}}$$

(iii) Since $\cos \theta = e$, the x-coordinate of P is ae .

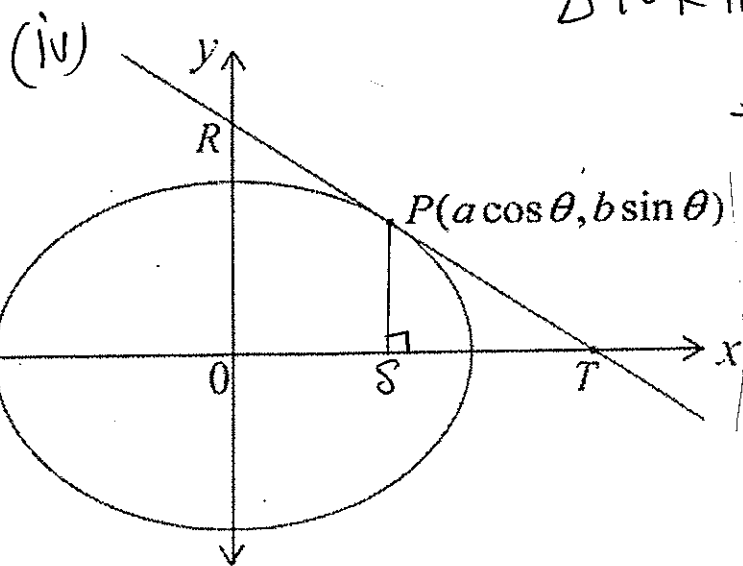
Coordinates of focus = $(ae, 0)$

Point P and focus S are on the vertical line

$$x = ae.$$

$\therefore$ the focal chord makes an angle of 90° with the x-axis

$\triangle TOR \parallel \triangle TSP$ (equiangular)



$$\frac{RT}{OT} = \frac{RP}{OS} \text{ (ratio of intercepts)}$$

$$\frac{RT}{\frac{a}{e}} = \frac{RP}{ae}$$

$$\frac{RT \times e}{a} = \frac{RP}{ae}$$

$$RP = \frac{RT \times e \times ae}{a}$$

$$\underline{\underline{RP = RT \times e^2}}$$

$$(b) (i) y = 16 - 4x$$

$$y = (4 - x)^2$$

Side of equilateral Δ

$$= 16 - 4x - (x - 4)^2$$

$$= 16 - 4x - (x^2 - 8x + 16)$$

$$= 16 - 4x - x^2 + 8x - 16$$

$$= 4x - x^2$$

Area of equilateral Δ

$$= \frac{1}{2} (4x - x^2)^2 \times \sin 60$$

$$= \frac{1}{2} x^2 (4 - x)^2 \times \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{4} x^2 (4 - x)^2$$

$$(ii) \Delta V = \frac{\sqrt{3}}{4} x^2 (4 - x)^2 \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^4 \frac{\sqrt{3}}{4} x^2 (4 - x)^2 \Delta x$$

$$= \frac{\sqrt{3}}{4} \int_0^4 x^2 (4 - x)^2 dx$$

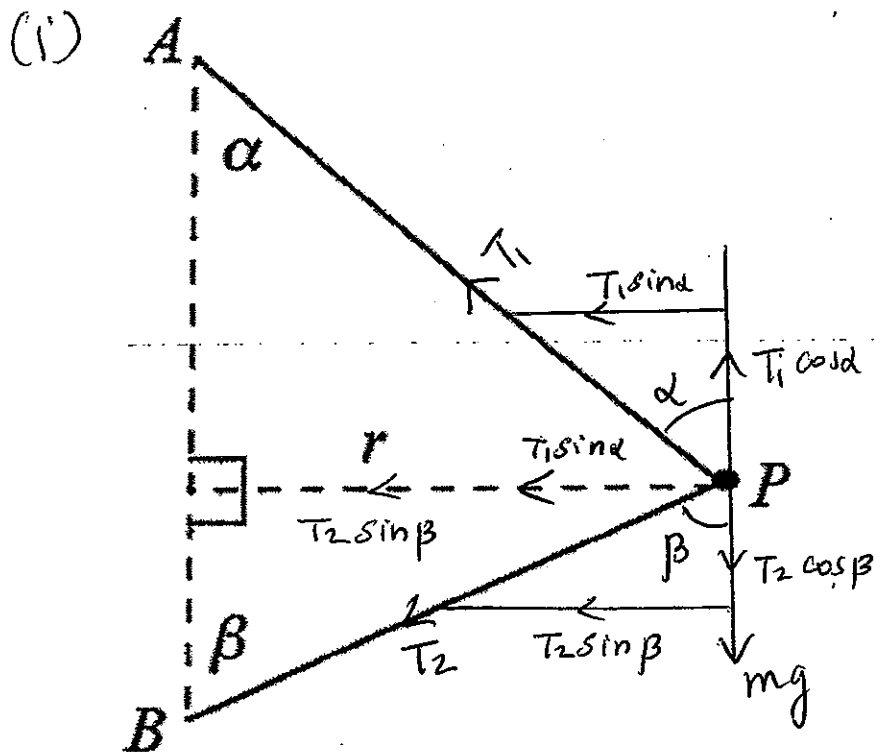
$$= \frac{\sqrt{3}}{4} \int_0^4 x^2 (16 - 8x + x^2) dx$$

$$= \frac{\sqrt{3}}{4} \int_0^4 (16x^2 - 8x^3 + x^4) dx$$

$$= \frac{\sqrt{3}}{4} \left[\frac{16x^3}{3} - \frac{8x^4}{4} + \frac{x^5}{5} \right]_0^4$$

$$= \frac{\sqrt{3}}{4} \left\{ \frac{16 \times 64}{3} - \frac{8 \times 256}{4} + \frac{1024}{5} \right\}$$

$$= \underline{\underline{14.78 \text{ U}^3}}$$



(ii) $T_1 \cos \alpha - T_2 \cos \beta = mg$ (since there is no vertical motion) — (1)

$T_1 \sin \alpha + T_2 \sin \beta = m r \omega^2$ (centripetal force is provided by $T_1 \sin \alpha$ and $T_2 \sin \beta$) — (2)

① $\times \sin \beta$

$T_1 \cos \alpha \sin \beta - T_2 \cos \beta \sin \beta = mg \sin \beta$ — (3)

② $\times \cos \beta$

$T_1 \sin \alpha \cos \beta + T_2 \sin \beta \cos \beta = m r \omega^2 \cos \beta$ — (4)

③ + ④ $T_1 (\sin \alpha \cos \beta + \cos \alpha \sin \beta) = m (r \omega^2 \cos \beta + g \sin \beta)$

$T_1 \times \sin (\alpha + \beta) = m (r \omega^2 \cos \beta + g \sin \beta)$

$T_1 = \frac{m (r \omega^2 \cos \beta + g \sin \beta)}{\sin (\alpha + \beta)}$

① $\times \sin \alpha$

$$T_1 \cos \alpha \sin \alpha - T_2 \cos \beta \sin \alpha = mg \sin \alpha \quad \text{--- (5)}$$

② $\times \cos \alpha$

$$T_1 \sin \alpha \cos \alpha + T_2 \sin \beta \cos \alpha = m r \omega^2 \cos \alpha \quad \text{--- (6)}$$

⑥ - ⑤

$$T_2 (\sin \beta \cos \alpha + \sin \alpha \cos \beta) = m (r \omega^2 \cos \alpha - g \sin \alpha)$$

$$T_2 \sin(\alpha + \beta) = m (r \omega^2 \cos \alpha - g \sin \alpha)$$

$$T_2 = \frac{m (r \omega^2 \cos \alpha - g \sin \alpha)}{\sin(\alpha + \beta)}$$

$$T_1 > 0 \quad \text{since} \quad m, r, \omega, \cos \beta, \sin \beta, g > 0$$

$$T_2 > 0 \implies r \omega^2 \cos \alpha - g \sin \alpha > 0$$

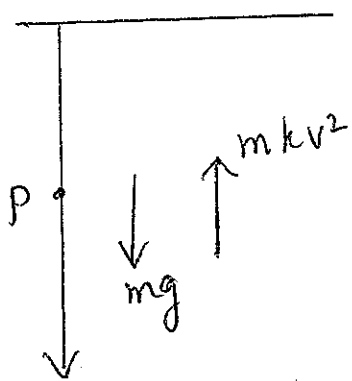
$$r \omega^2 \cos \alpha > g \sin \alpha$$

$$\omega^2 > \frac{g \sin \alpha}{r \cos \alpha}$$

$$\omega^2 > \frac{g \tan \alpha}{r}$$

$$\omega > \sqrt{\frac{g \tan \alpha}{r}}$$

(b)(i)



Equation of motion is
 $m\ddot{c} = mg - mkv^2$

$$\ddot{c} = g - kv^2$$

The terminal velocity occurs when the acceleration of the particle becomes zero.

$$g - kv^2 = 0$$

$$g = kv^2$$

$$v^2 = \frac{g}{k}$$

$$\underline{v = \sqrt{\frac{g}{k}}}$$

(ii) $\ddot{c} = g - kv^2$

substitute $\ddot{c} = v \frac{dv}{dc}$

$$v \frac{dv}{dc} = g - kv^2$$

$$v dv = (g - kv^2) dc$$

page 16

$$\frac{v dv}{g - kv^2} = dc$$

$$\int_0^s dc = \int_0^w \frac{v dv}{g - kv^2}$$

$$[c]_0^s = \int_0^w \frac{-2kv dv}{-2k(g - kv^2)}$$

$$s = \frac{-1}{2k} \int_0^w \frac{-2kv dv}{g - kv^2}$$

$$= \frac{-1}{2k} \left[\log(g - kv^2) \right]_0^w$$

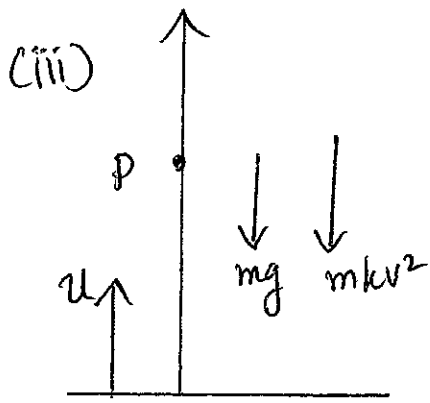
$$= \frac{-1}{2k} \left(\log(g - kw^2) - \log g \right)$$

$$= \frac{-1}{2k} \left(\log \left(\frac{g - kw^2}{g} \right) \right)$$

$$= \frac{-1}{2k} \log \left(1 - \frac{kw^2}{g} \right)$$

$$= \frac{-1}{2k} \log \left(1 - \frac{w^2}{v^2} \right)$$

$$\underline{\underline{=}} \text{ since } \frac{k}{g} = \frac{1}{v^2}$$



$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -g - kv^2$$

$$v \frac{dv}{dx} = -g - kv^2$$

$$v dv = -(g + kv^2) dx$$

$$\frac{v dv}{-(g + kv^2)} = dx$$

$$\int_0^H dx = \int_u^0 \frac{v dv}{-(g + kv^2)}$$

$$[x]_0^H = - \int_u^0 \frac{v dv}{g + kv^2}$$

$$= \int_0^u \frac{v dv}{g + kv^2}$$

$$= \int_0^u \frac{2kv dv}{2k(g + kv^2)}$$

page 17

$$H = \frac{1}{2k} \int_0^u \frac{2kv dv}{g + kv^2}$$

$$= \frac{1}{2k} \left[\log(g + kv^2) \right]_0^u$$

$$= \frac{1}{2k} \left(\log(g + ku^2) - \log g \right)$$

$$= \frac{1}{2k} \log \left(\frac{g + ku^2}{g} \right)$$

$$= \frac{1}{2k} \log \left(1 + \frac{ku^2}{g} \right)$$

$$= \frac{1}{2k} \log \left(1 + \frac{u^2}{v^2} \right)$$

$$\therefore \frac{k}{g} = \frac{1}{v^2}$$

(iii) The distances travelled by the particle in going up and coming down are the same $\therefore S = H$

$$-\frac{1}{2k} \log \left(1 - \frac{w^2}{v^2} \right) = \frac{1}{2k} \ln \left(1 + \frac{u^2}{v^2} \right)$$

$$\frac{1}{2k} \log \left(1 - \frac{w^2}{v^2} \right)^{-1} = \frac{1}{2k} \log \left(1 + \frac{u^2}{v^2} \right) \quad \text{page 18}$$

$$\left(1 - \frac{w^2}{v^2} \right)^{-1} = 1 + \frac{u^2}{v^2}$$

$$\left(\frac{v^2 - w^2}{v^2} \right)^{-1} = \frac{v^2 + u^2}{v^2}$$

$$\frac{1}{\left(\frac{v^2 - w^2}{v^2} \right)} = \frac{v^2 + u^2}{v^2}$$

$$\frac{v^2}{v^2 - w^2} = \frac{v^2 + u^2}{v^2}$$

$$v^4 = (v^2 - w^2)(v^2 + u^2)$$

$$v^4 = v^4 + v^2 u^2 - w^2 v^2 - w^2 u^2$$

$$0 = v^2 u^2 - w^2 v^2 - w^2 u^2$$

$$\div \text{ by } u^2 v^2 w^2$$

$$0 = \frac{v^2 u^2}{u^2 v^2 w^2} - \frac{w^2 v^2}{u^2 v^2 w^2} - \frac{w^2 u^2}{u^2 v^2 w^2}$$

$$0 = \frac{1}{w^2} - \frac{1}{u^2} - \frac{1}{v^2}$$

$$0 = w^{-2} - u^{-2} - v^{-2}$$

$$\underline{\underline{w^{-2} = u^{-2} + v^{-2}}}$$

Question 1b

(a) (i) $(y+1)^2 - x^2 = 1$

$2(y+1) \frac{dy}{dx} - 2x = 0$

$\frac{dy}{dx} = \frac{2x}{2(y+1)} = \frac{x}{1+y}$

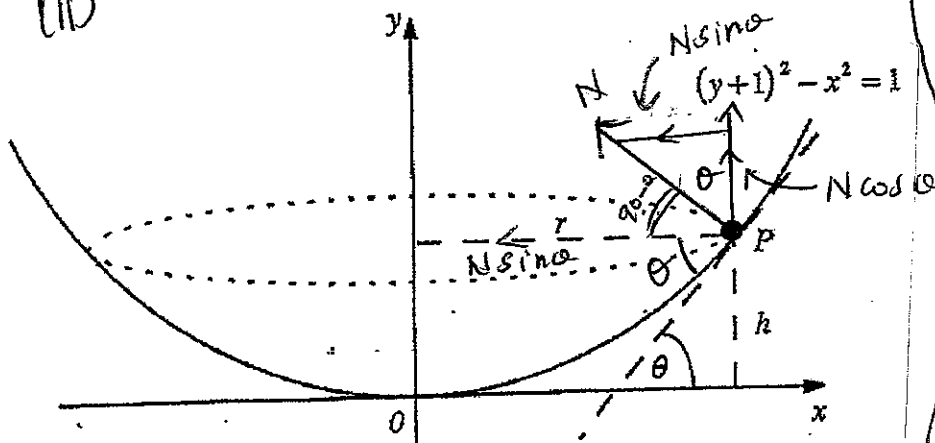
$\frac{dy}{dx}$ at (x_1, y_1)

$= \frac{x_1}{1+y_1}$ = gradient of the tangent at (x_1, y_1)

Also gradient = $\tan \theta$

$\therefore \tan \theta = \frac{x_1}{1+y_1}$

(ii)



Vertical component $N \cos \theta$ balances mg and horizontal component $N \sin \theta$ provides

Centripetal force

$N \cos \theta = mg$ — (1)

$N \sin \theta = mr\omega^2$ — (2)

(2) $\div$ (1)

$\frac{N \sin \theta}{N \cos \theta} = \frac{mr\omega^2}{mg}$

$\tan \theta = \frac{r\omega^2}{g}$

$\frac{x_1}{1+y_1} = \frac{r\omega^2}{g}$

$\therefore = P(r, h)$

$x_1 = r \quad y_1 = h$

$\frac{r}{1+h} = \frac{r\omega^2}{g}$

$\omega^2 = \frac{rg}{r(1+h)} = \frac{g}{1+h}$

(iv) From (1)

$N = \frac{mg}{\cos \theta} = mg \sec \theta$

$N = mg \sqrt{1 + \tan^2 \theta}$

page 20

$$N = mg \sqrt{1 + \frac{r^2}{(1+h)^2}}$$

But (r, h) is on the parabola

$$(y+1)^2 - x^2 = 1$$

$$(h+1)^2 - r^2 = 1$$

$$r^2 = (h+1)^2 - 1$$

$$\frac{r^2}{(1+h)^2} = 1 - \frac{1}{(1+h)^2}$$

$$N = mg \sqrt{1 + 1 - \frac{1}{(1+h)^2}}$$

$$= mg \sqrt{2 - \frac{1}{(1+h)^2}}$$

(v) linear speed $v = r\omega$

$$v^2 = r^2 \omega^2$$

$$\frac{3g}{2} = r^2 \times \frac{g}{1+h}$$

$$\frac{3}{2} = \frac{r^2}{1+h}$$

$$\frac{3}{2} = \frac{(h+1)^2 - 1}{1+h}$$

$$3(1+h) = 2(1+h)^2 - 2$$

$$2(1+h)^2 - 3(1+h) - 2 = 0$$

This is a quadratic in $1+h$

$$(-4, 1)$$

$$p+q = -4$$

$$p+q = -3$$

$$2(1+h)^2 - 4(1+h) + (1+h) - 2 = 0$$

$$2(1+h) [(1+h) - 2] + (1+h) - 2 = 0$$

$$[(1+h) - 2] [2(1+h) + 1] = 0$$

$$(h-1) (2+2h+1) = 0$$

$$(h-1) (2h+3) = 0$$

$$h = 1 \quad (\because h \text{ is positive})$$

$$N = mg \sqrt{2 - \frac{1}{(1+h)^2}}$$

$$= mg \sqrt{2 - \frac{1}{4}} = mg \sqrt{\frac{8-1}{4}}$$

$$= mg \sqrt{\frac{7}{4}} = \frac{1}{2} mg \sqrt{7}$$

(b) $g = 10 \text{ m/s}^2$, $r = 450 \text{ m}$

$\theta = \tan^{-1}\left(\frac{1}{5}\right)$ (ii)

$\tan \theta = \frac{1}{5}$

design speed

$V = \sqrt{rg \tan \theta}$

$= \sqrt{450 \times 10 \times \frac{1}{5}}$

$= 30 \text{ m/s}$

$= 108 \text{ km/h}$

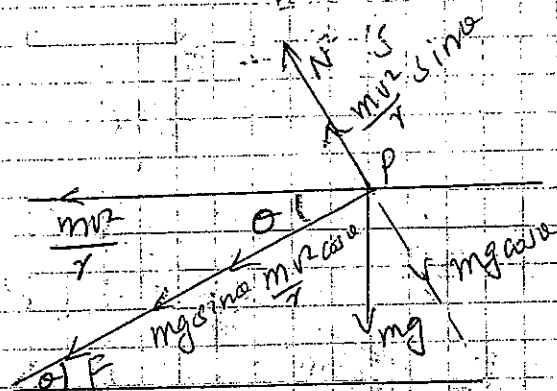
(i) When the vehicle travels with the design speed of the track, frictional force is zero.

126 km/h.

The vehicle is travelling at a speed higher than the design speed.

$126 \text{ km/h} = \frac{126000}{3600} \text{ m/s}$

$= 35 \text{ m/s}$



$\frac{mv^2}{r} \cos \theta = F + mg \sin \theta$

$F = mg \cos \theta \left(\frac{v^2}{rg} - \tan \theta \right)$

$= \frac{5000 \times 10 \times 5}{\sqrt{26}} \left(\frac{35 \times 35}{450 \times 10} - \frac{1}{5} \right)$

$= \frac{250000}{\sqrt{26}} \left(\frac{1225}{4500} - \frac{1}{5} \right)$

$= 3540.98$

$= 3541 \text{ N down track}$

